SAR Image Segmentation Based on Constrained Smoothing and Hierarchical Label Correction

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***Abstract*— Synthetic aperture radar (SAR) is widely used in the field of modern remote sensing due to its high resolution for a comparatively small antenna. However, there are still some difficulties in the processing of SAR images. In particular, accurate segmentation of small targets and image corners remains an important challenge, as these can easily be lost during conventional image smoothing and denoising methods. To address this, we propose an SAR image segmentation algorithm based on constrained smoothing and hierarchical label correction (CSHLC). First, a Canny algorithm is used to extract the edges of SAR images, and the Gaussian smoothing is performed on SAR images under edge constraints to achieve noise reduction so that the edges of small and big targets are well preserved. Second, a preliminary K-means clustering is conducted on the smoothing results, and then, a Markov random field (MRF) model is used on the clustering results ("original label" results), iteratively calculating a maximum likelihood set of pixel labels. Finally, through two label correction methods, pixel group counting comparison (PGCC) and gray similarity comparison (GSC), the labels of the MRF output are further checked and corrected to obtain final segmentation results. Compared with seven state-of-the-art algorithms, simulation results on both simulated SAR images and real SAR images show that the proposed CSHLC delivers higher accuracy while better retaining corners and small targets.**

***Index Terms*— Edge constrained smoothing (ECS), K-means, Markov random field (MRF), synthetic aperture radar (SAR), unsupervised segmentation.**

I. Introduction

IN SURFACE remote sensing analysis and processing, a variety of imaging devices can generate different types of remote sensing images, including optical images [1], infrared images [2], and radar images [3]. Synthetic aperture radar (SAR) is an active microwave remote sensing imaging device, with inherent advantages in spatial resolution [4], [5]. Meanwhile, because of the all-day and all-weather working characteristics [6], SAR systems are widely used in many common fields, such as hydrology, environmental monitoring, and geological exploration [7], [8]. However, the imaging reflection mechanism of SAR causes speckle noise in SAR images [9]. Speckle noise is multiplicative [10]. Therefore, it is impossible to use the classical methods of dealing with additive noise to reduce the noise of SAR images, which restricts the further understanding and processing of SAR images [11], [12].

So far, image segmentation, as an important step in image analysis, is still a challenging task [13]. Image segmentation divides an image into homogeneous, nonoverlapping regions of pixel labels, according to grayscale, texture, and other pixel features [14], [15]. Image segmentation often includes supervised segmentation [16], semisupervised segmentation [17], and unsupervised segmentation [18]. Many deep-learning models have been applied to supervised segmentation; however, these often need large data sets and training time. Recently, a boundary-weighted domain adaptive neural network (BOWDA-Net) [13] has demonstrated improved sensitivity to boundaries during segmentation. An advanced boundary-weighted transfer learning method is proposed to successfully deal with small data sets. In contrast to supervised segmentation methods, the unsupervised methods do not need to use labeled data that are expensive and difficult to collect [19]. Therefore, they can also achieve segmentation tasks well when there is little *a priori* information [20]. There are many methods in the conventional digital image segmentation literature, such as edge-based segmentation algorithms, region-based segmentation strategies, and threshold-based segmentation techniques [21]. However, these simple segmentation methods struggle to cope with SAR images that are polluted by speckle noise. In recent years, unsupervised SAR image segmentation papers have exploited a variety of methods, including active contour models [22], random field models [23], clustering-based methodologies [24], and watershed-based theories [25], amongst others.

Segmentation methods based on active contour models drive initialization curves close to target boundaries through

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energy functions [26] so that the curves can be used as target segmentation lines. These methods are robust to noise and have considerable advantages in complex target segmentation applications [27]. However, their shortcomings include computational complexity, slow processing speed, and difficulty with incorporating very fine-scale features [28]. Segmentation methods based on random field models represent the image segmentation as a probability problem and realize the segmentation by calculating the maximum likelihood set of labels for all pixels [29]. These methods frequently consume significant memory and computation time [30] because they need to iteratively optimize over the set of all possible pixel labels. Random field models have the advantage of utilizing local and global information simultaneously, so it is easy to obtain more accurate and refined segmentation results [31]. MRF models have also been extended to incorporate *a priori* knowledge of various kinds [32], [33]. Clustering-based methods are widely used in the field of image segmentation [34]. The K-means clustering algorithm is welcomed by many scholars due to its simplicity, concise implementation steps, and efficient execution efficiency [35]. However, the K-means algorithm is relatively sensitive to noise, so it commonly requires data smoothing prior to clustering [36]. Fuzzy c-means (FCM) clustering, using the fuzzy mathematics theory, determines the final data attribution according to membership [37]. Thus, compared with K-means that is a hard partition method, FCM can make better use of spatial information. Watershed-based segmentation methods, which mainly implement watershed transformation on edge strength maps to achieve initial segmentation [25], are usually closely related to region merging. Watershed transformation is a powerful tool in mathematical morphology. It is easy to operate and can construct closed segments on the image [38]. However, it often produces an oversegmentation phenomenon.

In order to ensure high execution efficiency, this article mainly considers using clustering methods to achieve SAR image segmentation. The main challenge of clustering-based segmentation is how to effectively reduce speckle noise in SAR images [3]. The smaller the influence of speckle noise on the clustering algorithm, the better will be the clustering results that can be obtained. Scholars suggest different solutions to noise reduction before clustering. Ji *et al.* [3] proposed a robust nonlocal fuzzy clustering algorithm with a between-cluster separation measure (NSFCM) for SAR image segmentation. The NSFCM [3] uses non-local mean filtering to reduce speckle noise in SAR images and then exploits the FCM with an improved objective function to obtain good segmentation results. Xiang *et al.* [39] proposed a kernel FCM algorithm with pixel intensity and location information (ILKFCM) for SAR image segmentation. The ILKFCM [39] realizes noise reduction by wavelet decomposition and then uses kernel clustering to measure the similarity of the objective function during the clustering iteration process to achieve segmentation. Moreover, a fast algorithm for SAR image segmentation based on key pixels (FKPFCM) [24] was proposed. The FKPFCM [24] uses certain rules to divide image pixels into key pixels and nonkey pixels and segments them, respectively, to achieve SAR image segmentation. The method utilizes a low-pass Gaussian filter to reduce the noise of SAR images in the early stage. Shang *et al.* [37] proposed a thumbnail-based hierarchical fuzzy clustering algorithm (THFCM) for SAR image segmentation. The THFCM [37] first narrows the original image to generate a thumbnail based on the gray value characteristics. The low-resolution thumbnail can achieve the purpose of suppressing noise. Next, the improved FCM is used to cluster the thumbnail, and the original SAR image can be effectively segmented according to the clustering results.

Filtering methods cannot completely restore noise-free SAR images, but they can assist with ensuring that different pixels of the same target tend to be homogeneous after smoothing [3]. Thus, the different pixels of the same target have a greater probability of being clustered together and marked as the same label. Hence, from the perspective of image segmentation, filtering and smoothing can reduce the noise of SAR images [24]. However, if the same filtering operation is performed on all areas of the entire image in the smoothing process, ordinary filters, such as mean filtering and Gaussian filtering, only can obtain effective noise reduction results in some areas. At target boundaries, these ordinary filters will inevitably lead to the degradation of target edges [40], which causes many pixels of target boundaries to be wrongly segmented in clustering results. Therefore, in order to effectively reduce noise while also preserving image detail, the characteristics of different areas should be considered in the image. In this article, a Gaussian smoothing method based on region growing and edge constraints is proposed. The method uses different smoothing operations according to different region characteristics, which preserves image edges, while also effectively reducing speckle noise in SAR images.

Most clustering-based SAR image segmentation algorithms finish segmentation after clustering. However, if the clustering results also can be processed effectively, the segmentation accuracy can be further improved. Shang *et al.* [41] proposed an SAR image segmentation algorithm using region smoothing and label correction (RSLC). The RSLC [41] uses a majority voting label correction method to further process the clustering results, and the experiment results show that postprocessing is effective. Inspired by RSLC, this article proposes three label correction methods as a hierarchical postprocessing chain.

Although the RSLC algorithm demonstrates excellent segmentation performance, like many SAR image segmentation algorithms, the algorithm still has some shortcomings. For RSLC, key remaining challenges include insufficient segmentation capabilities for small targets and image corners. Since the smoothing template used in the process of noise reduction has a certain width, when the width of the smoothed target is close to the template width, it is difficult to obtain an ideal smoothing effect, which causes the loss of some pixel values and even final segmentation errors. In order to solve the problem and achieve more accurate SAR image segmentation, an SAR image segmentation algorithm based on constrained smoothing and hierarchical label correction (CSHLC) is proposed in this article. First, Canny edge detection is utilized to extract the edges of the image following preliminarily smoothed, and the extracted edges are then used as constraints to perform edge constrained smoothing (ECS).

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Therefore, noise can be effectively reduced while still preserving the edges of both small and large targets. Second, the K-means algorithm is used to conduct preliminary clustering and segmentation on the smoothed image. Then, using the clustering results as initial label information, a Markov random field (MRF) model is used to iteratively compute the maximum probability set of pixel labels. This has the useful effect of correcting wrongly segmented corners and some small targets. Finally, pixel group counting comparison (PGCC) and gray similarity comparison (GSC) are used to retain the label information corrected by MRF and further check and correct other pixel labels so that final segmentation results are obtained.

The main contributions of the proposed algorithm are given as follows.

1) Image edges, extracted by the Canny algorithm, are used as constraints while performing Gaussian smoothing to enable appropriate smoothing of targets with different sizes. This enables large targets to be effectively smoothed and denoised without causing the edges of small targets to degenerate. Therefore, the information about small objects can be effectively preserved.
2) K-means clustering is used to obtain an initial segmentation result, which is then iteratively refined using an MRF. This step is useful for correcting wrongly marked labels of many small targets and corners in the image.
3) Using the MRF output pixel labels as an initialization, PGCC and GSC are used to correct the remaining wrongly segmented pixels.

The remainder of this article is structured as follows. In Section II, the proposed algorithm is introduced in detail. In Section III, the experiment settings are given, and the experimental results are presented and analyzed. Section IV provides concluding remarks.

II. Proposed Method

The proposed algorithm in this article comprises three main steps. The first step is image smoothing based on region growing and edge constraints. The second step is corner label correction based on an MRF model. The third step is label correction based on PGCC and GSC.

A. Image Smoothing Based on Edge Constraints

Speckle noise in SAR images can seriously compromise segmentation results. Therefore, it is often necessary to denoise SAR images before segmentation. For image segmentation tasks, the requirement for noise reduction is relatively low. There is no need to restore the gray value of the ground truth. It is only necessary to make pixels with similar characteristics have more similar pixel values after noise reduction, after which clustering the pixels realizes a correct image segmentation. Image smoothing is a simple and effective way to meet the above requirement. Because the pixel values of different regions have different distribution characteristics, each region needs to be analyzed individually to achieve more effective smoothing. The smoothing characteristics of different regions of the image are shown in Fig. 1.

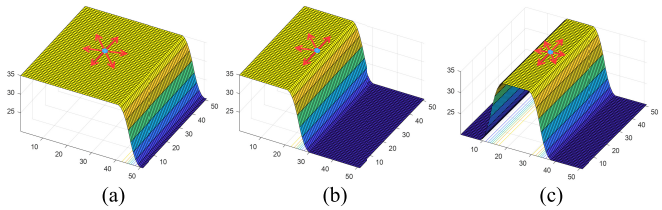


Fig. 1. Smoothing characteristics of different regions in the image: (a) smoothing in the wide region; (b) smoothing in the region whose one side closes to edges; and (c) smoothing in the region whose multiple sides close to edges.

As can be seen from Fig. 1, there are three different types of smoothing operations, according to the characteristics of different regions. In Fig. 1, the value of the vertical axis represents the gray value. The direction and length of the red arrows indicate the direction of the filtering and the width of the filter kernel, respectively. Therefore, different combinations of red arrows indicate different filtering operations. Fig. 1(a) illustrates the most common type of regions, which have homogenous surroundings. These require an isotropic smoothing method. Fig. 1(b) illustrates regions constrained by edges on one side. Only the pixels with the same features (i.e., the gray values of these pixels are close) should be smoothed. Such regions usually exist at the junctions of different targets. The more the targets in the image, the more this type of regions appears. Fig. 1(c) shows a class of regions constrained by edges on multiple sides. There should be fewer pixels involved in smoothing at the directions close to edges, and more pixels involved in smoothing at the directions away from edges. These regions are often located inside small targets.

Our proposed algorithm attempts to select and apply the most appropriate choice of smoothing, based on the distances from the centroid (center pixel) of the smoothing region to the surrounding edges. If the centroid is far away from the edges in all directions, the corresponding smoothing region belongs to the type shown in Fig. 1(a). The isotropic smoothing template is suitable for this type of smoothing region. If one direction of the centroid is close to an edge, and other directions are far away from edges, the type of the corresponding smoothing region is shown in Fig. 1(b). The smoothing template with a one-sided edge constraint is suitable for this kind of smoothing region. If the centroid is close to the edges in multiple directions, then it belongs to the type shown in Fig. 1(c). A smoothing template with multiside edge constraints is suitable for this type of smoothing region. Since these smoothing operations all need to use image edges for measurement, the edge information needs to be extracted first. The extracted edges should cover all real edges of the image, and there should be as few false edges as possible. The real edges usually occur at the boundaries of real target objects. In contrast, false edges are edges that are detected by mistake due to noise, and they commonly appear inside the boundaries of targets. A small quantity of false edges does not affect smoothing results, but the fewer the false edges, the better the smoothing effect.

Because of the severe speckle noise in SAR images, traditional edge detection methods cannot be used directly on

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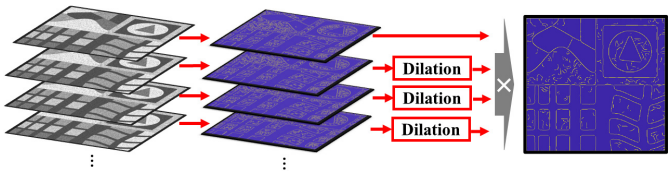


Fig. 2. Image edge extraction process.

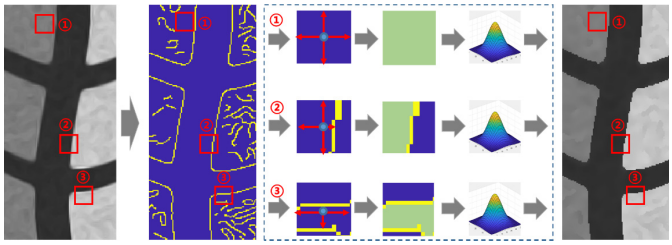


Fig. 3. Edge constrained smoothing process.

SAR images. The directional Gaussian smoothing that we used in [41] can reduce the influence of noise to a certain extent, so the smoothing method is used to perform preliminary smoothing on the original image. Then, a Canny algorithm is used to extract the edges of the image. Note that the parameters of the Canny algorithm are adaptive [42]. The image edge extraction process is shown in Fig. 2.

As can be seen from Fig. 2, in order to make the extracted edges have fewer false edges, the Canny algorithm is used to detect image edges after each preliminary smoothing. Since the image edges often appear 1-pixel offset, the extracted edges are dilated by one-pixel width. If there are N_1 times of preliminary smoothing, N_1 times of edge extraction and $N_1 - 1$ times of edge dilation are needed. The final image edges are obtained by multiplying $N_1 - 1$ dilation edges and the N_1^{th} extracted image edges. Because real image edges are not lost during the smoothing process and the positions of false edges extracted each time are different, the above operations can remove some false edges in the image. The calculation formulas for edge extraction are given as follows:

$$E_{\text{tmp}}^{(k)} = E_{\text{thin}}^{(k)} * A, \quad k \in [1, N_1 - 1] \tag{1}$$

$$E_{\text{thick}}^{(k)}(x, y) = \begin{cases} 1, & \text{if } E_{\text{tmp}}^{(k)}(x, y)! = 0 \\ 0, & \text{else} \end{cases} \tag{2}$$

$$E_{\text{sm}} = E_{\text{thin}}^{N_1} \cdot \prod_{k=1}^{N_1 - 1} E_{\text{thick}}^{(k)} \tag{3}$$

where $E_{\text{thin}}^{(k)}$ represents the edges detected on the basis of the kth smoothing, and $*$ indicates the convolution operation. A, representing the structuring element of dilation, is a 3×3 all-one matrix. N_1 is the number of preliminary smoothing. $E_{\text{thick}}^{(k)}$ represents the dilated edges on the basis of the kth edge detection. E_{sm} is the final image edge for ECS.

Next, ECS is conducted to reduce noise while retaining image details. Specifically, taking the obtained edges as constraints, a Gaussian filter is used to slide sequentially for smoothing different areas in the image. For each sliding position, the region as large as the smoothing window is intercepted, and the connected components of the center pixel in the region are obtained by using a region-growing algorithm. Finally, the product of the connected components and the Gaussian function is used as the final filtering result. The ECS process is shown in Fig. 3.

Fig. 3 shows the image without smoothing, the image edges, the smoothing process, and the smoothing result from left to right, respectively. The three red rectangular boxes shown in Fig. 3 correspond to three types of regions: no edge constrained region, one-sided edge constrained region, and multisided edge constrained region. It can be seen from Fig. 3 that, for the unconstrained region (region number 1: top row), connected components of the center pixel include all pixels in the rectangle after the region growing. This corresponds to complete the Gaussian smoothing. For the one-sided edge constrained region (region number 2: middle row), the growing area of the center pixel is merely part of the sliding window. Therefore, only the pixels on one side of the box are utilized for Gaussian smoothing (see the third part in Fig. 3). For the multisided edge constrained region (region number 3: bottom row), the growing area of the center pixel just has the middle part of the sliding window, so only the pixels in the middle of the box are used for Gaussian smoothing (see the third part in Fig. 3). From the final smoothing result, it is clear that the difference between gray values in the homogeneous region becomes smaller, and the edges of targets are clearly sharpened. The formulas of the ECS process are expressed as follows:

$$G_{\text{ES}}(x, y) = \exp\left(-\frac{x^2 + y^2}{2\sigma_g^2}\right) \tag{4}$$

$$Q_{x,y} = \eta_4\{W_{\text{gs}}, E_{\text{sm}}, X_{\text{esm}}(x, y)\} \tag{5}$$

$$X_{\text{sm}}(x, y) = \frac{1}{H} G_{\text{ES}} \cdot Q_{x, y} \tag{6}$$

where G_{ES} is the template of Gaussian smoothing and σ_g represents the standard deviation of the Gaussian function. $Q_{x,y}$ represents the region-growing result at coordinates (x, y). $\eta_4\{\cdot\}$ indicates the four-neighbor region growth operation, and W_{gs} represents the diameter of the region-growing window. X_{esm} is the image after preliminary smoothing, and X_{sm} is the image after ECS. H represents the normalization term.

B. Corner Label Correction Based on Markov Random Field

Note that the labels mentioned in this article are actually pseudolabels, which are different from those used in image classification, because no physical interpretation can be assigned to the pseudoclasses. In this article, the pixels segmented together are marked with the same number. For better visualization, different numbers correspond to different colors. These numbers and colors used only for better processing and description are called labels.

Section II-A described the smoothing procedure for reducing the influence of speckle noise in SAR images by using edge extraction and ECS. This process results in different pixels of the same target being assigned more similar intensity values. Next, K-means clustering is used to cluster the

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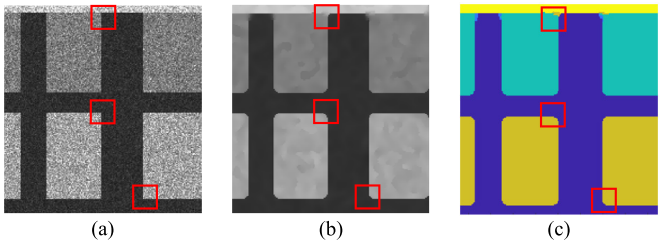


Fig. 4. Information loss of image corner pixels: (a) original SAR image; (b) smoothed image; and (c) K-means clustering result.

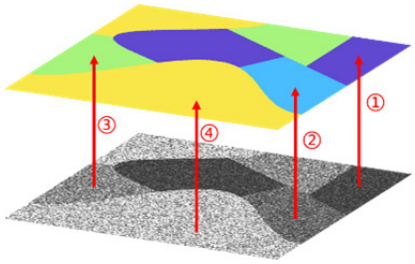


Fig. 5. Image labeling task.

smoothed image pixels, and a relatively good segmentation result can be obtained. However, due to the scale problem, many corner pixels are damaged during the smoothing process and cannot be segmented correctly, as shown in Fig. 4.

Fig.4 illustrates how corner pixels become lost during smoothing operations [see Fig. 4(b)] and subsequent clustering [see Fig. 4(c)]. As shown in Fig. 4(b), the corresponding corners in smoothing results are dulled, and the corner pixel information is lost, which causes these corners to be wrongly segmented during clustering [see Fig. 4(c)].

The reason for the problem is given as follows. Although small smoothing templates with a width of 5 or 3 can be used, there are often corners with smaller widths in images. The pixel values of these smaller corners are gradually averaged during the smoothing process, which results in the loss of corner information. Failing to detect corners during segmentation will destroy the details of the image and cause a decrease in segmentation accuracy. Therefore, it is necessary to further process these corners.

Segmentation is the process of partitioning an image into a set of nonintersecting regions such that each region is homogeneous but the union of no two adjacent regions is homogenous. Image segmentation can be regarded as a pixel labeling task (see Fig. 5).

When performing segmentation on noisy images, we assign each pixel a label that is most likely to be correct given uncertain information. Assuming that the input image is $X = \{x_{ij} |\ x_{ij} \in R^{m\times n}\}$ and the output of pixel label set is $B = \{b_{ij} |\ b_{ij} \in N^{m\times n},\ b_{ij} > 0\}$, the label of pixel x_{ij} is b_{ij}. Assuming that the given label set is $W = \{w_i |\ w_i \in N^C,\ w_i > 0\}$, $b_{ij} \in W$, and $C \leq m \times n$. The *a posteriori* probability $P(b_{ij}|x_{ij})$ is expressed by the Bayesian formula as follows:

$$P\left(b_{ij}|x_{ij}\right) = \frac{p\left(x_{ij}|b_{ij}\right)P\left(b_{ij}\right)}{p\left(x_{ij}\right)} \tag{7}$$

where $p(x_{ij}|b_{ij})$ is the class-conditional probability density for label b_{ij}. $P(b_{ij})$ is the *a priori* probability for label b_{ij}, and $p(x_{ij})$ is the prior probability density of the observed pixel intensity, given distribution of all pixel intensities in the observed image.

Since the distribution of the input image is completely known, for input image X, solving the maximum posterior probability problem can be transformed into the problem of optimizing the set of pixel labels B that maximize the likelihood

$$f(X, B) = \arg\max_{B}(P(B)p(X|B)). \tag{8}$$

Since the mutual influence relationship between pixels has conditional independence, the pixel label field can be regarded as an MRF [32], [33]. According to the Hammersley–Clifford theorem, MRF and Gibbs random field are equivalent [17]. Therefore, the term $P(B)$ can be evaluated by utilizing Gibbs distribution.

After logarithmic processing, the optimization problem can be transformed into the formulas as follows:

$$f(X, B) = \arg\min_{B}\{\varphi_1(X,\ B) + \varphi_2(B)\} \tag{9}$$

$$\varphi_1(X, B) = \sum_{w_i\in W}\left(\ln\left(\sqrt{2\pi}\sigma_{w_i}\right) + \frac{\left(X - \bar{\mu}_{w_i}\right)^2}{2\sigma_{w_i}^2}\right) \tag{10}$$

$$\varphi_2(B) = \sum_{z\in Z}\sum_{b_{ij}\in B} V_z\left(b_{ij}\right) \tag{11}$$

$$V_z\left(b_{ij}\right) = \begin{cases} -\beta, & b_{ij} = b_{gf} \\ +\beta, & b_{ij} \neq b_{gf} \quad b_{ij}, b_{gf} \in B \end{cases} \tag{12}$$

where $\varphi_1(X,\ B)$ represents the logarithm form of the class-conditional probability density $p(X|B)$. σ_{w_i} and $\bar{\mu}_{w_i}$, respectively, are standard deviation and mean value of the gray values of all pixels marked with label w_i. $\varphi_2(B)$ represents the energy of the Gibbs random field. $V_Z(b_{ij})$ represents the potential energy of potential cluster z for label b_{ij}, and Z represents the set of potential energy clusters. In this article, the second-order potential energy clusters are used. β is the coupling coefficient, and its value is a constant. b_{ij} and b_{gf} are the labels of two adjacent pixels.

The result of K-means clustering is used as *a priori* label information, and the above optimization problem is solved iteratively so that the maximum likelihood set of pixel labels for the input image can be obtained. Since ECS can suppress speckle noise very well, K-means clustering can segment most of the pixels in the image correctly. Therefore, most pixel labels remain unchanged after MRF, while the wrongly marked corner labels are gradually corrected under the influence of neighborhood information. The corner label correction process is shown in Fig. 6.

Fig. 6(a) shows the corner map of the original image marked with pixel values. Fig. 6(b) shows the corner label map of the K-means clustering result, where the question marks are wrongly marked pixels. Fig. 6(c) shows the iterative process of correcting corner labels. As can be seen from the pixel values of the original image, the corner pixels with question marks should be marked with a yellow label. However, the smoothing process causes the loss of corner pixel information so that the corner pixels are wrongly labeled in clustering result [see Fig. 6(b)]. Since the labels of pixels are more affected by

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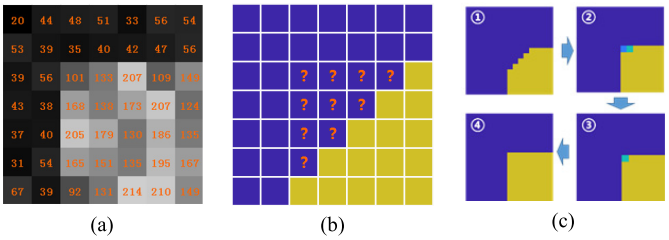


Fig. 6. Image corner label correction process: (a) original image corner; (b) K-means corner labeling result; and (c) iterative MRF label correction process.

their neighboring pixels, the corner pixel labels are gradually corrected during the MRF iteration process [see Fig. 6(c)].

C. Label Correction Based on Pixel Group and Zone Growth

MRF iteration process, to achieve the overall optimization, corrects wrongly marked corner labels, while incorrectly changing a small number of pixel labels. These wrongly corrected pixels are called extreme points. After MRF, the pixels of a corner usually enter a pixel group, including a large number of pixels marked as the same label, whereas the extreme point pixels will generate new pixel groups independently. According to the difference, the wrong labels of extreme points can be corrected by the PGCC label correction method. The so-called pixel group is a group of pixels connected and gathered together through four or eight neighbors. This article uses four neighbors to get pixel groups. The PGCC label correction process is shown in Fig. 7.

As shown in Fig. 7, the difference map can be got first according to the original label map and MRF label correction map. The original label map is obtained by K-means clustering. Then, the number of pixels in each pixel group of the MRF label map is counted, which is called pixel group counting (PGC). Similarly, PGC is implemented on the difference map, that is, to calculate the number of pixels per pixel group in the difference map. Next, this article compares the counting result of each pixel point in the MRF label map with that of the corresponding pixel point in the difference map. For any pixel point, if the counting results of the MRF label map and difference map are the same, the label of the pixel point in the MRF label map will be corrected to the label of the corresponding pixel point in the original label map. If the counting results are different, the label of the current pixel point in the MRF label map will be retained. The formulas for the PGCC label correction process are expressed as follows:

$$F(x,y)=\begin{cases}1, & \text{if } L_{\mathrm{KM}}(x,y)\neq L_{\mathrm{MRF}}(x,y)\\ 0, & \text{else}\end{cases} \tag{13}$$

$$L_{\mathrm{PGCC}}(x,y)=\begin{cases}L_{\mathrm{KM}}(x,y), & \text{if } F(x,y)=1\\ & \text{and } N_F(x,y)=N_{\mathrm{MRF}}(x,y)\\ L_{\mathrm{MRF}}(x,y), & \text{else}\end{cases} \tag{14}$$

where L_{KM} and L_{MRF}, respectively, represent the original label map and MRF label map. F is the difference map of L_{KM} and L_{MRF}. N_F is the counting result of pixel groups of the difference map, and N_{MRF} is the counting result of pixel groups of the MRF label map. L_{PGCC} represents the label correction result of PGCC.

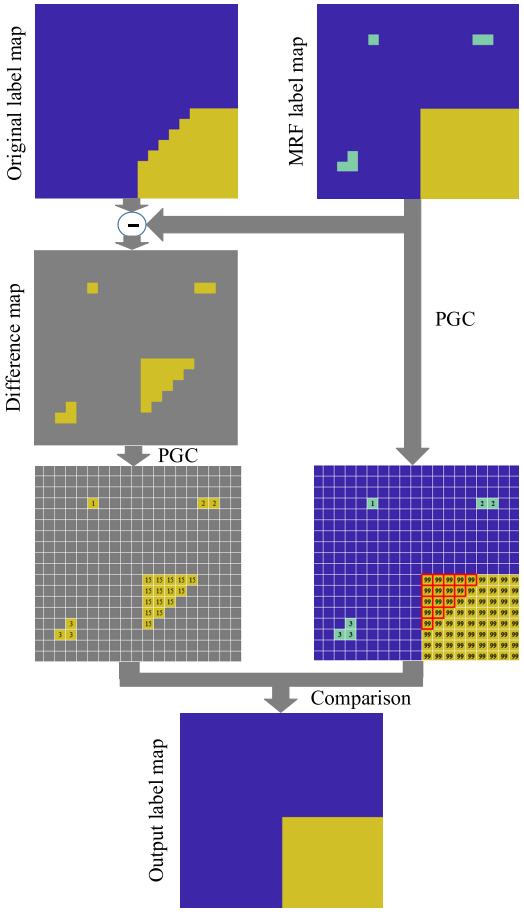


Fig. 7. Label correction process of PGCC.

After PGCC label correction, the labels of wrongly corrected extreme points are restored while effectively retaining corrected corner labels. Although the current segmentation result is better than the clustering result, in fact, there may still be some wrongly labeled pixels. These mislabeled pixels are generated by K-means clustering and are not corrected by iterative MRF. In order to correct the remaining wrong pixel labels, it is necessary to further process the MRF label results.

The label correction method in [41] based on region growth and majority voting has achieved a good label correction effect. By improving this method, we propose a label correction method called GSC that is based on region growth and gray value comparison. GSC needs to use the edge information of the image. In order to retain corrected corner information, this article needs to use not only the previous edges obtained in Section II-A but also the edges extracted from the MRF label map. Similarly, the Canny algorithm is first used to extract edges of the MRF label map. Then, both the edges are fused. The edge fusion formula is given as follows:

$$E_{\mathrm{sc}}(x,y)=\begin{cases}1, & \text{if } E_{\mathrm{sm}}(x,y)=E_{\mathrm{MRF}}(x,y)\\ 0, & \text{else}\end{cases} \tag{15}$$

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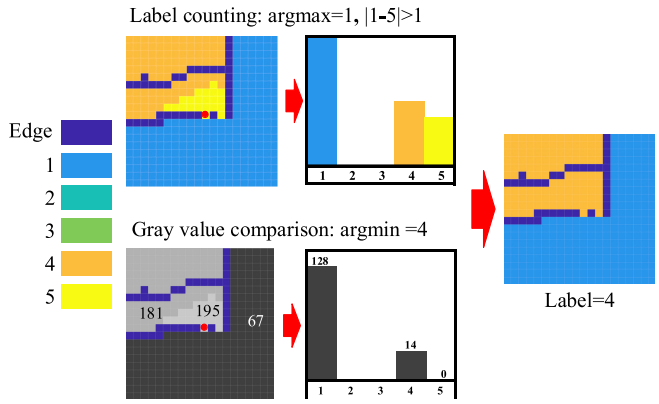


Fig. 8. Label correction process of GSC.

where E_{sc} is the fused edge result, which is used to conduct ESC label correction. E_{sm} is the edges obtained in Section II-A. E_{MRF} represents the edge result of the MRF label map.

The ESC label correction process is shown in Fig. 8.

Fig. 8 shows the label correction process of a region with edge fracture. The red dot represents the current pixel to be corrected, namely, the center pixel. According to the majority voting label correction method [41], the label of the center pixel should be corrected to 1, but this value is quite different from the original label value 5. It can also be seen from the gray image in Fig. 8 that the gray value of the center pixel is 195, whereas the gray mean of the pixels labeled with 1 is 67. It is obvious that the two gray values should not belong to the same target. By gray value comparison, it can be found that the difference between the gray mean of the pixels labeled with 4 and the gray value of the center pixel is smaller. Therefore, the label of the center pixel should be corrected to 4. As shown in the gray image, the gray mean of the region with label 4 is 181, which is more likely to belong to the same type of target as the center pixel with the gray value of 195.

The specific reason is that, under the influence of noise, the gray values of the image will fluctuate within a certain range. However, the gray values of a small number of pixels deviate from the gray mean values of the real targets that they belong to but are closer to the average gray values of other targets. The targets with different gray values are marked in order from small to large when marking labels, which makes the wrong pixel labels in the image mainly occur between adjacent pixels. Therefore, the absolute value of the difference between the real label value and the wrong label value is usually not more than 1. Meanwhile, at edge fracture regions, the region-growing algorithm may search for the labels of different targets, and these labels are quite different from the center pixel label. Therefore, it is suitable to correct the center pixel label based on the similarity of gray values. The GSC label correction process is expressed by the following formula:

$$L_{MAX}(x, y) = \arg\max\left\{\eta_4\left[W_{gw}, L_{PGCC}(x, y)\right]\right\} \tag{16}$$

$$L_{GC}(x, y) = \arg\min\left\{\bar{\mu}_{L_{PGCC}(x,y)} - \bar{\mu}_{\eta_4}\right\} \tag{17}$$

$$L_{GSC}(x, y) = \begin{cases} L_{MAX}(x, y), & \text{if } \|L_{MAX}(x, y) - L_{PGCC} \times (x, y)\| \le 1 \\ L_{GC}(x, y), & \text{else} \end{cases} \tag{18}$$

where $\eta_4[\cdot]$ represents region-growing operation by four neighbors, and W_{gw} is the width of the region-growing window. L_{PGCC} is the label result obtained by PGCC. L_{MAX} is the label result obtained by majority voting label correction [41]. $\bar{\mu}_{L_{PGCC}(x,y)}$ represents the gray mean of the pixel group that $L_{PGCC}(x, y)$ belongs to. $L_{GC}(x, y)$ is the label of pixel point (x, y) after gray value comparison. $\bar{\mu}_{\eta_4}$ is the gray mean of each pixel group adjacent to the growing region with center pixel point (x, y). $L_{GSC}(x, y)$ represents the label correction result obtained by GSC at pixel point (x, y).

Finally, the two comparison correction results are fused according to the edges result obtained in Section II-A, and the fusion formulas are given as follows:

$$S(x, y) = \begin{cases} L_{PGCC}(x, y), & \text{if } E_{sm}(x, y) \neq 0 \\ L_{GSC}(x, y), & \text{else} \end{cases} \tag{19}$$

where S represents the segmentation result map. The definitions of $L_{PGCC}(x, y)$ and $L_{GSC}(x, y)$ are the same as above.

In order to obtain a better segmentation effect, the MRF algorithm and the PGCC label correction method can be used again to further improve the segmentation accuracy. The pseudocode of the proposed algorithm is shown in Algorithm 1.

Algorithm 1 *CSHLC*: SAR Image Segmentation Based on CSHLC

Require: X = original SAR image
Require: C = class number
Require: T = iteration number of MRF
1. X_{esm} = PreSmooth (X) ▷ preliminary smoothing
2. E_{sm} = EdgeExtraction (X) ▷ edge extraction
3. X_{sm} = EdgeConstrSm (X_{esm}, E_{sm}) ▷ edge constrained smoothing
4. L_{KM} = Kmeans (X_{sm}, C) ▷ K-means clustering
5. L_{MRF} = MRF (L_{KM}, T) ▷ corner label correction
6. L_{PGCC} = PGCC (L_{KM}, L_{MRF}) ▷ pixel group counting comparison
7. L_{GSC} = GSC (L_{PGCC}, E_{sm}, L_{MRF}) ▷ gray similarity comparison
8. S = Fusion (L_{PGCC}, L_{GSC}) ▷ correction result fusion
9. S_{MRF} = MRF (S)
10. R = PGCC (S, S_{MRF})
11. **Return** R ▷ final segmentation result

III. Experiment Settings and Result Analysis

In this section, to verify the effectiveness of the proposed algorithm for SAR segmentation, the results obtained by CSHLC are compared with those of five clustering-based algorithms, one MRF-based algorithm, and one watershed-based algorithm. The five clustering-based comparison algorithms

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include ILKFCM [39], NSFCM [3], incorporating adaptive local information into fuzzy clustering (ALFCM) [36], FKPFCM [24], and THFCM [37]. In addition, a fast, weighted CRF algorithm based on a two-step superpixel generation (FWCRF) [23] is the MRF-based comparison algorithm, and the region-merging method with texture pattern attention (RMTBA) [25] is the watershed-based comparison algorithm. The software MATLAB2017b is used to perform simulation experiments, and the computer is configured with Intel Core i7-7700 3.60 GHz, 16-GB RAM, and Win10 64-bit operating system.

A. Experimental Images

In the experiments, three sets of simulated SAR images that can be available from *https://github.com/Meng306/simulated-SAR-images.git* and one set of real SAR images are segmented. It has to be said that the preprocessing of raw SAR data is a very complex and difficult task. The preprocessing chain includes map projection [43], radiometric correction [44], [45], geometric correction [46], incidence angle correction [47], and so on. Incidence angle correction is also very laborious because, for certain target types, it is plausible but universally impossible in practice. This will lead to oversegmentation of, e.g., water (seas and lakes) depending on the wave spectrum and viewing geometry. However, the real SAR images used in this article have been preprocessed. The simulated SAR images are obtained by adding speckle noise to the original pollution-free artificial synthetic images, as shown in Fig. 9.

Fig. 9(a) shows the original clean image of SI1with the size of 256×256. There are four types of targets with gray values of 50, 100, 150, and 200 in SI1, respectively. Fig. 9(d), (g), and (j) shows the two-, four-, and six-look SI1 simulated SAR images. Fig. 9(b) shows the original clean image of SI2. Fig. 9(e), (h), and (k) shows the two-, four-, and six-look SI2 simulated SAR images, respectively. SI2 is a 384×384 image composed of four types of targets with gray values of 50, 100, 150, and 200, respectively. Fig. 9(c) shows the original clean image of SI3 with the size of 512×512. There are five types of targets with gray values of 50, 90, 130, 170, and 210 in SI3, respectively. Fig. 9(f), (i), and (l) shows the two-, four-, and six-look SI3 simulated SAR images.

In addition, the first group of simulation images is mainly composed of straight lines, including some corners, and can test the ability of algorithms to segment straight lines and corners. The second group of simulation images is mainly made up of curves, with some small targets, and can test the ability of algorithms to segment curves and small targets. The third group of simulation images contains straight lines and curves and has more targets. They can test the ability of algorithms to segment complex images.

Simulation experiments are also conducted on three real SAR images called Noerdlinger, Maricopa, and Traunstein, respectively, which are shown in Fig. 10.

Fig. 10(a) shows the first real SAR image, which was taken in the middle of the Swabian Jura mountains in southwestern Germany, called Noerdlinger. The size of Noerdlinger is 256×256. It can be divided into four different types of

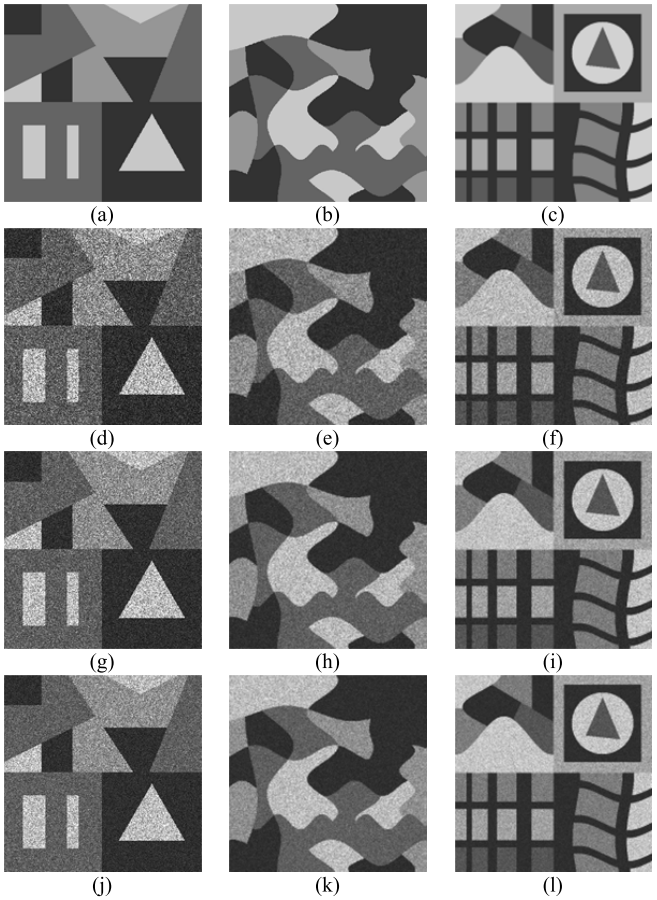


Fig. 9. Simulated SAR images: (a) SI1; (b) SI2; (c) SI3; (d), (g), and (j) two-, four-, and six-look SI1 simulated SAR images; (e), (h), and (k) two-, four-, and six-look SI2 simulated SAR images; and (f), (i), and (l) two-, four-, and six-look SI3 simulated SAR images.

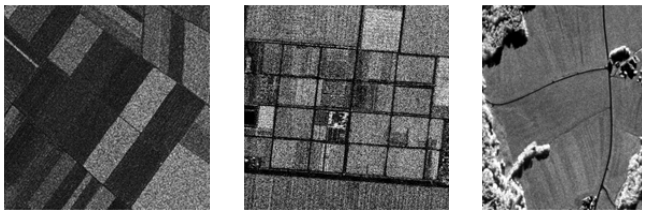


Fig. 10. Real SAR images: (a) Noerdlinger image; (b) Maricopa image; and (c) Traunstein image.

targets, with a resolution of 1 m. The second real SAR image in Fig.10(b) was captured at the Maricopa Agricultural Center near Arizona in the United States, called Maricopa. It can be divided into four types of targets. The image size is 350×350, and its resolution is 1 m. Fig. 10(c) shows the third real SAR image, called Traunstein. This image that can be segmented into three types of targets was imaged in Bavaria, Germany. The Traunstein image has a size of 1001×779, and its resolution is 1 m. Among the three real images, Noerdlinger and Maricopa have large noise, and Traunstein is less affected by noise.

B. Evaluation Indexes

In order to quantitatively evaluate the effectiveness of the proposed algorithm, segmentation accuracy SA and kappa

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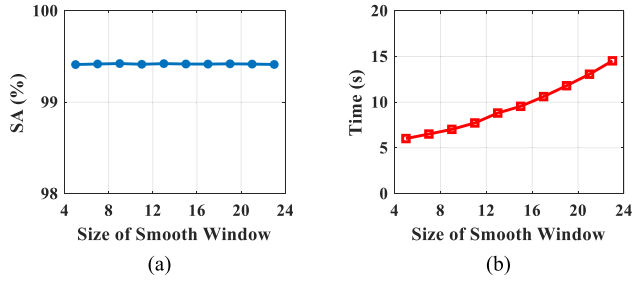


Fig. 11. Impact of parameter W_{ES} on the results: (a) impact on SA and (b) impact on the running time.

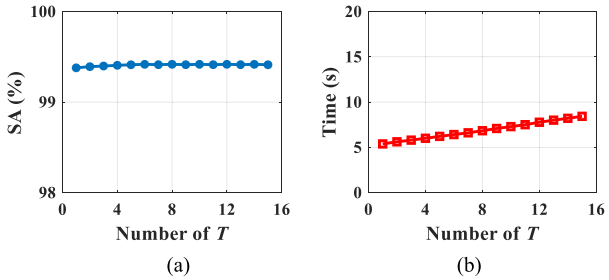


Fig. 12. Impact of parameter T on the results: (a) impact on SA and (b) impact on the running time.

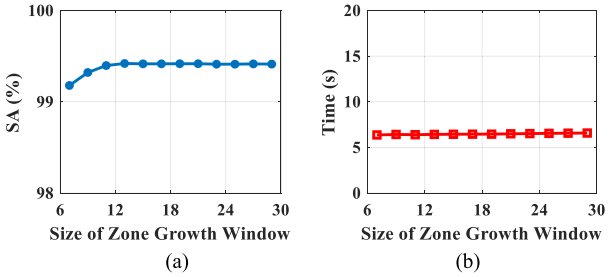


Fig. 13. Impact of parameter W_{gw} on the results: (a) impact on SA and (b) impact on the running time.

coefficient KC are used as evaluation indicators. The calculation formula of SA is given as follows:

$$\mathrm{SA} = \sum_{i=1}^{c} R_i \cap T_i \Big/ \sum_{i=1}^{c} T_i \tag{20}$$

where c represents the maximum value of all label values. R_i is the set of pixels that are labeled with i in the segmentation result, and T_i is the set of pixels that are labeled with i in the ground truth.

KC is used to measure the consistency between the segmentation result and the corresponding ground truth. The calculation formula is given as follows:

$$\mathrm{KC} = \left(\sum_{i=1}^{c} P_{ii} - \sum_{i=1}^{c} P_{i\cdot} P_{\cdot i} \right) \Big/ \left(1 - \sum_{i=1}^{c} P_{i\cdot} P_{\cdot i} \right) \tag{21}$$

where $P_{i\cdot} = \sum_{j=1}^{c} P_{ij}$ and $P_{\cdot j} = \sum_{i=1}^{c} P_{ij}$. $P_{\cdot j}$ and $P_{\cdot i}$ have the same meaning and form. P_{ij} represents the ratio of the number of pixels with label i in the segmentation result to the number of pixels with label j in the ground truth. Therefore, P_{ii} represents the ratio of the number of pixels with label i in the segmentation result to the number of pixels with label i in the ground truth.

The value of SA is between 0 and 1, and KC is usually the same. The closer the two evaluation indexes are to 1, the better the segmentation effect.

C. Parameter Analysis

In the experiments, there are several important parameters that need to be set according to the experimental content. They are the width of the ECS template W_{ES}, the K value of K-means, the number of MRF iterations T, and the region-growing window width W_{gw} used in the GSC label correction process. Among them, the relatively fixed parameter K is set as the number of the types of targets in the image. For the other three parameters, taking the two-look SI3 image as an example and using SA and running time as indicators illustrate the effects of them on the segmentation results.

First, the parameter W_{ES} is analyzed, and T and W_{gw} are set to 6 and 19, respectively. W_{ES} is set to 10 values from 5 to 23 with a step of 2. The test results are shown in Fig. 11.

From Fig. 11(a), it can be seen that the ECS template width W_{ES} has a relatively small impact on SA. Under different values of W_{ES}, the SA of the results is always maintained above 99%. When W_{ES} is 9, SA achieves the maximum value. It can be seen from Fig. 11(b) that, as the parameter W_{ES} increases from 5 to 23, the running time continues to increase. The main reason is that the increase in W_{ES} will increase the time of ECS. Considering time and SA at the same time, the width of the ECS template can be selected from 5 to 11.

Next, the parameter T is tested, and its value is set from 1 to 15 with a step of 1. During the test, W_{ES} and W_{gw} are set to 7 and 19, respectively. The experimental results are shown in Fig. 12.

It can be seen from Fig. 12(a) that, as the parameter T increases, the SA first increases slightly, and then, it basically remains steady. The reason is that, as the number of iterations increases, the mismarked corner labels are continuously corrected, and the SA is also increasing. When the number of corrected corner labels in the image reaches a certain degree, the increase in T will no longer increase the SA. As can be seen from Fig. 12(b), with the increase in T, the time consumption is basically on the rise. The main reason is that the increase in T will increase the time consumption of the MRF corner label correction process. Considering SA and time comprehensively, the parameter T can be set between 5 and 7.

Finally, the parameter W_{gw} is tested, and its value is set from 5 to 29 with a step of 2. Meanwhile, the values of W_{ES} and T are set to 6 and 7, respectively. The test results are shown in Fig. 13.

It can be seen from Fig. 13(a) that, as the parameter W_{gw} increases from 5 to 29, the SA increases at first, then decreases slightly, and finally remains basically unchanged. The main reason is that, with the increase in W_{gw}, the label correction region is also enlarging, which makes the SA increase. However, if the W_{gw} is too large, the correct labels will be marked incorrectly and, thus, drop the SA. It can be

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TABLE I
EXPERIMENTAL RESULTS ON SI1 SIMULATE IMAGES

Algorithm	2-look		4-look		6-look	
	SA (%)	*KC*	*SA* (%)	*KC*	*SA* (%)	*KC*
ILKFCM [39]	93.02	0.901	93.35	0.906	93.23	0.904
NSFCM [3]	93.19	0.903	94.38	0.920	95.18	0.931
ALFCM [36]	95.49	0.936	97.78	0.968	98.54	0.979
FKPFCM [24]	95.89	0.942	96.81	0.955	96.24	0.947
FWCRF [23]	92.68	0.897	97.64	0.966	97.27	0.961
THFCM [37]	95.60	0.938	96.28	0.947	96.55	0.951
RMTPA[25]	96.34	0.948	96.59	0.951	97.03	0.958
CSHLC	**99.14**	**0.988**	**99.61**	**0.994**	**99.75**	**0.996**

seen from Fig. 13(b) that the parameter W_{gw} has very little effect on time consumption. Therefore, the value of W_{gw} can be selected from 11 to 21.

D. Results and Analysis on Simulated and Real SAR Images

1) Results and Analysis of SI1: The two-, four-, and six-look SI1 simulated SAR images are segmented, and the segmentation accuracy SA and kappa coefficient KC of each algorithm are obtained, as shown in Table I.

From the experimental results in Table I, it can be seen that the SA of each algorithm is maintained above 90%, indicating that CSHLC and comparison algorithms all have a good segmentation effect for the set of images. For the two-look SI1 image, FWCRF has the lowest SA, whereas, for the four- and six-look SI1 images, FWCRF has the third highest SA, second only to CSHLC and ALFCM. In addition, the lower the look of the image, the higher the degree of noise pollution, so FWCRF has a better processing effect on weakly noisy SAR images. Under different noise levels, the SA of CSHLC and comparison algorithms except FWCRF is relatively stable, which shows that they are more robust to noise. From the experimental results, the proposed algorithm has the highest SA and KC, which shows the effectiveness of CSHLC for SAR image segmentation. Fig. 14 shows the segmentation result of each algorithm for the two-look SI1 image.

In Fig. 14, FWRCF and ALFCM retain the edge and corner information in the image better, but there are more noise and poor regional consistency in their results [see Fig. 14(e) and (g)], which leads to a decrease in the SA of both the algorithms. NSFCM and ILKFCM have less noise in their results, but there are more wrong segmentation points at the target edges [see Fig. 14(h) and (i)]. These edge errors mainly result from edge degradation caused by image smoothing. The segmentation edges in the results of THFCM and FKPFCM are very rough. There are many wrongly segmented corners in both the results [see Fig. 14(d) and (f)]. RMTPA has good regional consistency and flat edges in its result, but the corner information is seriously lost, and some edges are not well preserved [see Fig. 14(c)]. The proposed algorithm preserves the corner information well, and the edges are relatively smooth in its result [see Fig. 14(b)], which shows the effectiveness of CSHLC for SAR image segmentation.

2) Results and Analysis of SI2: The two-, four-, and six-look SI2 simulated SAR images are segmented, and the segmentation accuracy SA and kappa coefficient KC of each algorithm are obtained, as shown in Table II.

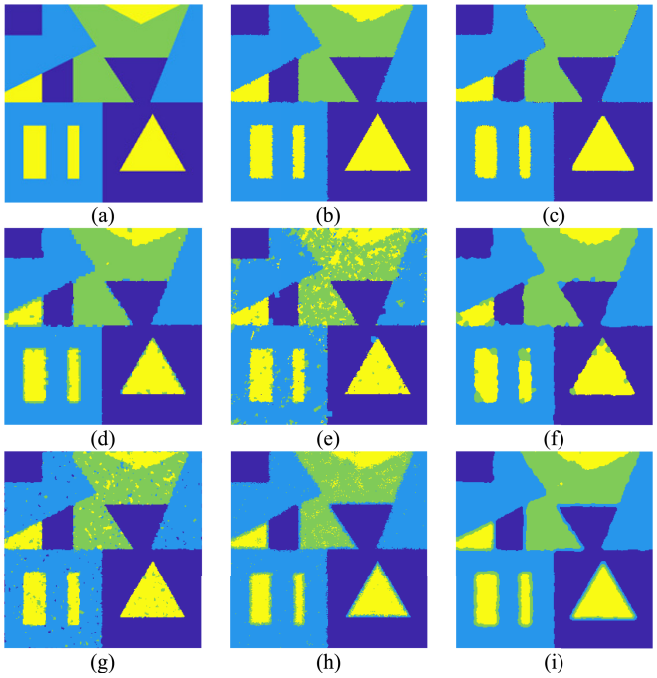


Fig. 14. Segmentation results on the two-look SI1 simulated image: (a) ground truth; (b) CSHLC; (c) RMTPA [25]; (d) THFCM [37]; (e) FWCRF [23]; (f) FKPFCM [24]; (g) ALFCM [36]; (h) NSFCM [3]; and (i) ILKFCM [39].

TABLE II
EXPERIMENTAL RESULTS ON SI2 SIMULATE IMAGES

Algorithm	2-look		4-look		6-look	
	SA (%)	*KC*	*SA* (%)	*KC*	*SA* (%)	*KC*
ILKFCM [39]	94.36	0.922	94.63	0.926	94.59	0.926
NSFCM [3]	93.90	0.916	95.21	0.934	95.71	0.941
ALFCM [36]	93.90	0.916	97.37	0.964	98.33	0.977
FKPFCM [24]	96.87	0.957	97.55	0.966	97.49	0.965
FWCRF [23]	94.37	0.922	97.84	0.970	98.84	0.984
THFCM [37]	96.01	0.945	97.00	0.959	96.86	0.957
RMTPA[25]	93.77	0.914	98.67	0.982	98.79	0.983
CSHLC	**99.41**	**0.992**	**99.70**	**0.996**	**99.80**	**0.997**

As can be seen from Table II, ILKFCM, NSFCM, FKPFCM, THFCM, and CSHLC all achieve stable and high SA for the SI2 image set, which shows that these algorithms have relatively high segmentation performance and good robustness to noise. However, the SA of ALFCM, FWCRF, and RMTPA varies greatly under different noise levels, indicating that these algorithms are more sensitive to noise. It can be seen from the result data that CSHLC is significantly higher than each comparison algorithm in terms of SA and KC, which shows the effectiveness of the algorithm for SAR image segmentation. Fig. 15 shows the segmentation result of each algorithm for the two-look SI2 image.

From Fig. 15, it can be seen intuitively that, for the results of FWCRF and ALFCM, there are a large number of wrongly segmented pixels inside the targets and at the segmentation boundaries [see Fig. 15(e) and (g)], so they are not satisfactory.

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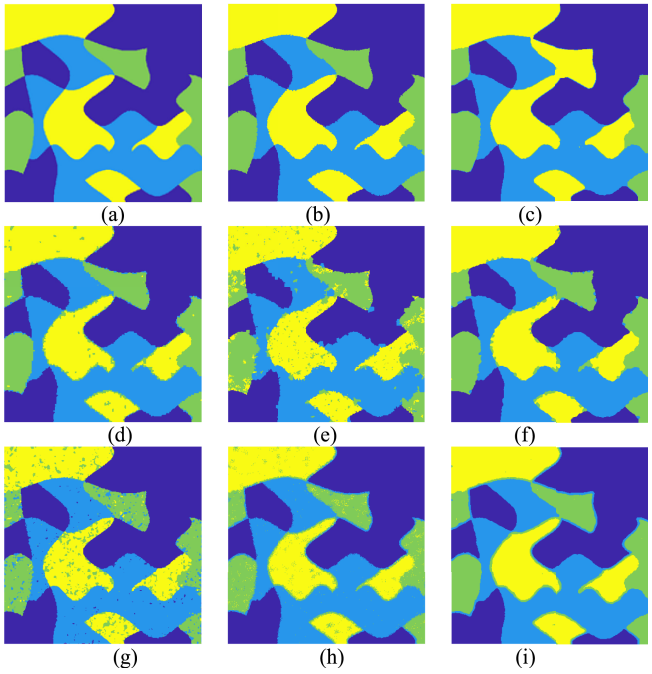


Fig. 15. Segmentation results on the two-look SI2 simulated image: (a) ground truth; (b) CSHLC; (c) RMTPA [25]; (d) THFCM [37]; (e) FWCRF [23]; (f) FKPFCM [24]; (g) ALFCM [36]; (h) NSFCM [3]; and (i) ILKFCM [39].

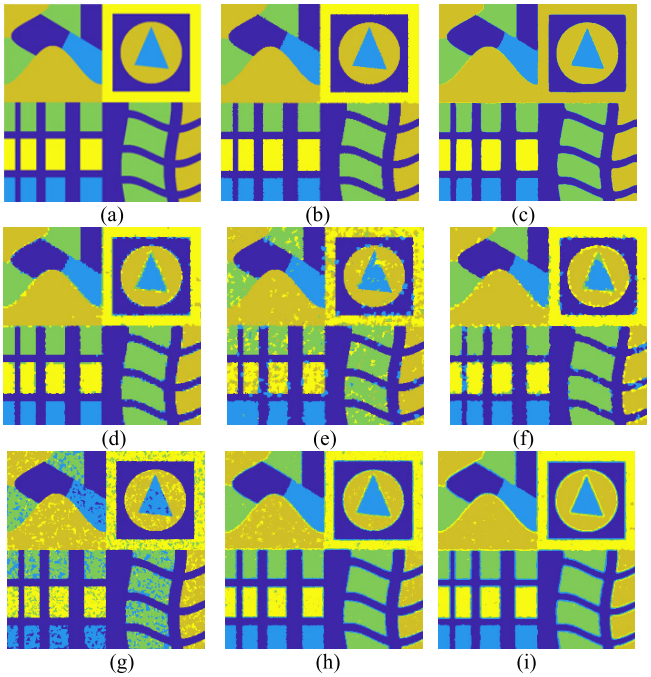


Fig. 16. Segmentation results on the two-look SI3 simulated image: (a) ground truth; (b) CSHLC; (c) RMTPA [25]; (d) THFCM [37]; (e) FWCRF [23]; (f) FKPFCM [24]; (g) ALFCM [36]; (h) NSFCM [3]; and (i) ILKFCM [39].

TABLE III
EXPERIMENTAL RESULTS ON SI3 SIMULATE IMAGES

Algorithm	2-look		4-look		6-look	
	SA (%)	*KC*	*SA* (%)	*KC*	*SA* (%)	*KC*
ILKFCM [39]	88.68	0.853	89.00	0.858	89.09	0.859
NSFCM [3]	89.64	0.864	91.43	0.887	92.94	0.907
ALFCM [36]	85.31	0.806	96.13	0.949	97.19	0.963
FKPFCM [24]	92.76	0.905	94.30	0.925	94.43	0.927
FWCRF [23]	87.42	0.835	92.28	0.899	97.67	0.969
THFCM [37]	94.07	0.938	94.58	0.929	94.80	0.932
RMTPA[25]	87.74	0.838	87.84	0.839	96.85	0.959
CSHLC	**99.45**	**0.992**	**99.75**	**0.997**	**99.87**	**0.998**

The results of THFCM, FKPFCM, and NSFCM have less noise and are cleaner overall, but there are many wrongly segmented pixels at the target edges, and the yellow tail-shaped small target in the image is not segmented correctly [see Fig.15(d), (f), and (h)]. For ILKFCM, there is no obvious noise in the result, but the target edges are blurred [see Fig. 15(i)]. For RMTPA, there is good regional consistency in the result, but some edges are obviously missing [see Fig. 15(c)]. Similarly, the yellow tail-shaped small target is segmented wrongly in both the results [see Fig. 15(c) and (i)]. Compared with the segmentation results, it can be seen that the proposed algorithm can correctly segment all large targets while completely segmenting the yellow tail-shaped small target in the image.

3) Results and Analysis of SI3: The two-, four-, and six-look SI3 simulated SAR images are segmented, and the segmentation accuracy SA and kappa coefficient KC of each algorithm are obtained, as shown in Table III.

The SI3 image set has more targets, and its size is also larger, so it is more difficult to segmentation. From the result data in Table III, it can be seen that, compared with the result data of SI1 and SI2 image sets, the SA of each comparison algorithm for the SI3 image set is significantly lower. Meanwhile, the SA of ALFCM, FWCRF, and PMTPA varies greatly under different noise intensities. It shows that the segmentation performance of these algorithms is affected by image complexity and noise. With the increase in the noise level, the SA of ILKFCM, NSFCM, FKPFCM, and THFCM is relatively stable, indicating that these algorithms are not sensitive to noise but sensitive to image complexity. In contrast, CSHLC has better segmentation performance and robustness to noise. Although the segmentation difficulty of the SI3 image set is obviously improved, the proposed algorithm still maintains the highest SA and KC under different noise levels, which shows that it has obvious advantages for SAR image segmentation. Fig. 16 shows the segmentation result of each algorithm on the two-look SI3 image.

As can be seen from Fig. 16, FWCRF and ALFCM have very serious noise in their results [see Fig. 16(e) and (g)], which is the reason why both the algorithms have the lowest SA on the two-look SI3 image. The results of NSFCM and ILKFCM have less noise inside the targets, but there are more mismarked pixels at the target edges [see Fig. 16(h) and (i)], which shows that both the algorithms have a good denoising effect on homogeneous regions but cannot effectively retain the edge information of the image. RMTPA, THFCM, and FKPFCM have good regional consistency in their result, but there are a large number of wrongly segmented corners

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TABLE IV
SEGMENTATION RESULTS ON REAL SAR IMAGES

Algorithm	Noerdlinger		Maricopa		Traunstein	
	SA (%)	*KC*	*SA* (%)	*KC*	*SA* (%)	*KC*
ILKFCM [39]	87.82	0.814	76.95	0.672	**90.46**	0.744
NSFCM [3]	90.36	0.849	79.66	0.707	88.26	0.677
ALFCM [36]	92.67	0.884	70.05	0.594	87.41	0.712
FKPFCM [24]	88.68	0.826	78.39	0.682	90.13	0.727
FWCRF [23]	80.09	0.679	70.94	0.593	90.02	**0.753**
THFCM [37]	91.10	0.862	80.60	0.716	89.00	0.692
RMTPA[25]	90.57	0.853	77.18	0.664	89.84	0.728
CSHLC	**96.38**	**0.943**	**91.57**	**0.878**	90.41	0.739

[see Fig. 16(c), (d), and (f)]. In addition, some edge information in the RMTPA result is not effectively preserved [see Fig. 16(c)]. Compared with the segmentation result of each algorithm, it can be seen that CSHLC can accurately segment each target whether it is small or large while completely retaining the corner structure of the triangles and quadrilaterals in the image. It shows that the proposed algorithm obviously has better overall segmentation ability and detail processing ability.

4) Results and Analysis of Real Images: The Noerdlinger, Maricopa, and Traunstein images are segmented. The segmentation accuracy SA and kappa coefficient KC of each algorithm are obtained for each real image, as shown in Table IV.

It can be seen from Table IV that, for the smaller Noerdlinger image, FWCRF has the lowest SA, and the SA of ILKFCM, NSFCM, FKPFCM, and RMTPA is slightly lower than that of ALFCM and THFCM, whereas CSHLC achieves the highest SA. Because the Maricopa image is more severely affected by noise and has many small targets, the SA of the comparison algorithms is not very high, and that of ALFCM and FWCRF is the lowest. The SA of CSHLC is much higher than that of all the comparison algorithms. The main reason is that the proposed CSHLC effectively retains the edge and corner information of the image and has better segmentation capability for small targets. For the Traunstein image with less noise, the SA of ALFCM and the KC of NSFCM are the lowest. ILKFCM achieves the highest SA with the value being 90.46%, whereas the SA of CSHLC is 90.41%, ranking second. The SA of both algorithms only differs by 0.05%. Compared with other comparison algorithms, it shows that both algorithms have better segmentation capabilities for weakly noisy images. The KC of CSHLC is 0.739, second only to FWCRF and ILKFCM with the KC value being 0.753 and 0.744, respectively. Compared with other comparison algorithms, it shows that the three algorithms have a more balanced ability to segment different targets. The segmentation results of the Noerdlinger are shown in Fig. 17.

In Fig. 17, FWCRF not only cannot correctly segment the large targets but also wrongly segment the small targets obviously [see Fig. 17(e)], which is the reason why it has the lowest SA. There is a lot of noise in the ALFCM result [see Fig. 17(g)], so it is not satisfactory. THFCM, NSFCM, and ILKFCM all have good segmentation results with less noise, and the small targets in the image are correctly segmented,

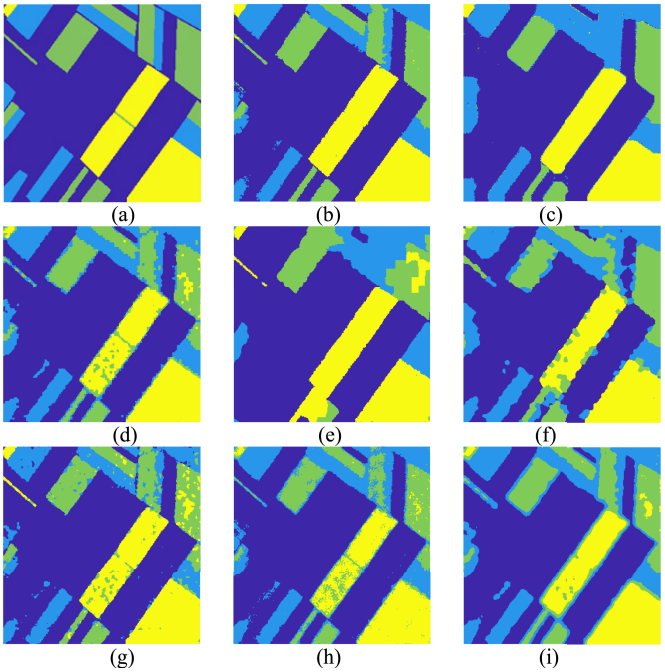


Fig. 17. Segmentation results on the Noerdlinger image: (a) ground truth; (b) CSHLC; (c) RMTPA [25]; (d) THFCM [37]; (e) FWCRF [23]; (f) FKPFCM [24]; (g) ALFCM [36]; (h) NSFCM [3]; and (i) ILKFCM [39].

but there are some wrongly segmented points at edges [see Fig. 17(d), (h), and (i)], which leads to decrease in SA. RMTPA and FKPFCM have many wrongly segmented pixel blocks in their results [see Fig. 17(c) and (f)]. Compared with the segmentation results, it can be seen that the result of CSHLC is clean and tidy, without obvious noise. The proposed algorithm has relatively smooth edges and better preserves the corners in the segmentation result. However, it still has a small flaw, which is that the green dividing line in the yellow region in the middle of the image is not well preserved. The main reason is that this small target has only one-pixel width, which is significantly smaller than the minimum smoothing template width. We will make further improvements in future work.

The segmentation results of the Maricopa image are shown in Fig. 18. As shown in Fig. 18, the segmentation results of FWCRF and ALFCM are scattered with a large number of wrongly segmented pixels [see Fig. 18(e) and (g)], which is the main reason why the SA of them is significantly lower than that of other algorithms. THFCM, NSFCM, and ILKFCM can segment most of the targets well in the image [see Fig. 18(d), (h), and (i)]. The segmentation effect of the three algorithms is better than that of FWCRF and ALFCM but is worse than that of the proposed CSHLC. The edge information is obviously missing in the results of RMTPA and FKPFCM [see Fig. 18(c) and (f)], which indicates that both the algorithms have a poor ability to segment small targets and edges. There is basically no visible noise in the segmentation result of CSHLC, and the interior of the targets is clean and unified [see Fig. 18(b)]. The proposed algorithm can accurately segment the large and small targets in the image, and its segmentation result has clear and visible segmentation

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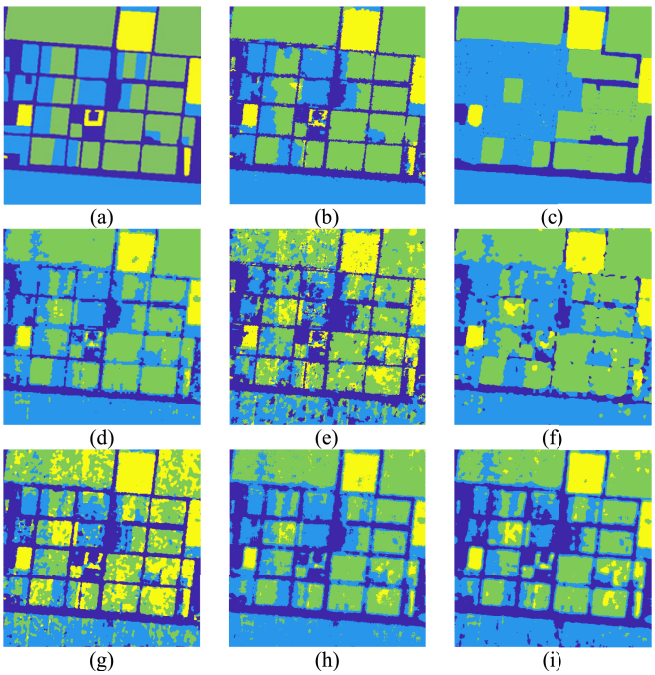


Fig. 18. Segmentation results on the Maricopa image: (a) ground truth; (b) CSHLC; (c) RMTPA [25]; (d) THFCM [37]; (e) FWCRF [23]; (f) FKPFCM [24]; (g) ALFCM [36]; (h) NSFCM [3]; and (i) ILKFCM [39].

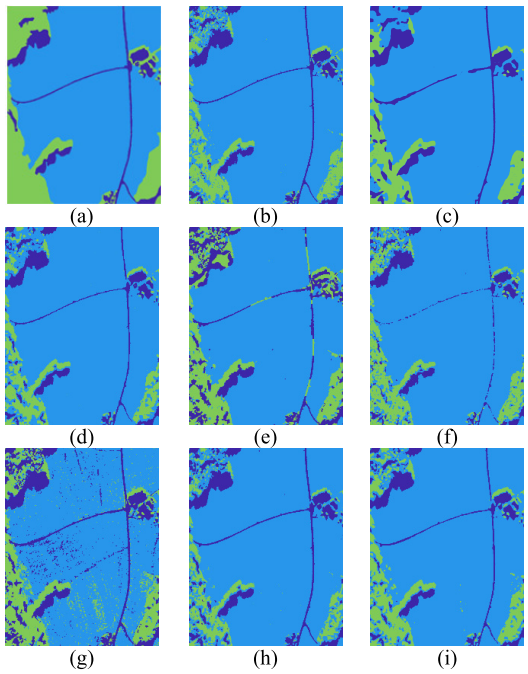


Fig. 19. Segmentation results on the Traunstein image: (a) ground truth; (b) CSHLC; (c) RMTPA [25]; (d) THFCM [37]; (e) FWCRF [23]; (f) FKPFCM [24]; (g) ALFCM [36]; (h) NSFCM [3]; and (i) ILKFCM [39].

boundaries, so the segmentation performance of CSHLC is obviously better than that of all the comparison algorithms.

The segmentation results of the Traunstein image are shown in Fig. 19.

In Fig. 19, ALFCM has more obviously wrongly segmented pixels in the result [see Fig. 19(g)], so the segmentation effect is slightly worse. For FWCRF, the banded targets in the middle of the image have many wrongly segmented pixels [see Fig. 19(e)], indicating that the ability of the algorithm to segment small targets is limited. ILKFCM has the highest SA, but, visually, it, as well as RMTPA, THFCM, FKPCM, NSFCM, and CSHLC, has almost the same segmentation result. Because the Traunstein image is less affected by noise and its size is larger with a small number of targets, RMTPA, THFCM, FKPFCM, NSFCM, and ILKFCM that are sensitive to image complexity also can obtain relatively good segmentation results. Meanwhile, it shows that the proposed CSHLC has good segmentation ability for low-noise and low-complexity SAR images.

In this section, we have implemented many experiments on real and simulated SAR images. Experimental results show that the proposed CSHLC is effective for SAR image segmentation. Furthermore, CSHLC could also be adapted for use on some multichannel SAR imagery (e.g., multiresolution SAR imagery). One straightforward way is to apply the proposed algorithm to each channel separately and then combining the separate channel results.

TABLE V
SMOOTHING EFFECTS ON SIMULATED AND REAL SAR IMAGES

Image	ADF [48]	AMF [49]	ECS
2-look SI1	77.75	89.83	**96.56**
4-look SI1	89.19	93.85	**97.97**
6-look SI1	95.46	96.25	**98.92**
2-look SI2	75.12	85.28	**95.36**
4-look SI2	84.75	91.41	**97.04**
6-look SI2	94.19	94.79	**98.60**
2-look SI3	65.57	77.32	**95.82**
4-look SI3	79.85	85.96	**97.35**
6-look SI3	88.67	90.79	**98.78**
Noerdlinger	85.07	86.91	**90.74**
Maricopa	61.90	69.21	**82.98**
Traunstein	82.88	87.31	**89.21**

E. Effectiveness of Edge Constrained Smoothing

In order to verify the effectiveness of ECS, the smoothing methods that can preserve edge information well, namely, anisotropic diffusion filtering (ADF) [48] and anisotropic median filtering (AMF) [49], and ECS are compared. For image segmentation tasks, the requirement for noise reduction is relatively low. There is no need to restore the gray value of the ground truth. It is only necessary to make pixels with similar characteristics have more similar pixel values after processing. In the experiment, the image is smoothed first; then, the K-means is used to segment the smoothed image; and finally, SA is calculated as an index to evaluate the performance of smoothing methods in the segmentation task. The SA of each smoothing method for every image is shown in Table V.

It can be seen from Table V that the smoothing effect of AMF is better than that of ADF. However, the smoothing effect of the ECS has absolute advantages compared with the other two smoothing methods. It shows the effectiveness of the proposed smoothing method, ECS, in image segmentation tasks.

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TABLE VI
RUNNING TIME ON REAL SAR IMAGES

Algorithm	Noerdlinger	Maricopa	Traunstein
	Time (s)	Time (s)	Time (s)
ILKFCM [39]	91.25	170.01	914.59
NSFCM [3]	12.32	23.53	146.94
ALFCM [36]	49.36	74.25	1095.96
FKPFCM [24]	8.03	15.06	373.02
FWCRF [23]	3.73	6.42	50.15
THFCM [37]	4.34	9.29	49.00
RMTPA[25]	10.99	239.39	6116.29
CSHLC	**1.58**	**3.48**	**38.61**

F. Comparison of Running Time

In order to verify the efficiency of the proposed CSHLC, the running times of eight algorithms are compared. In this section, the real SAR images that have more differences in size, number of targets, and noise level are taken as the experimental objects. The running time of every algorithm for each real image is shown in Table VI.

Table VI shows that ILKFCM and ALFCM have a relatively large time consumption. For RMTPA, with the increase in the image size, the time consumption increases greatly, which shows that the image size has a great impact on the running time of this algorithm. NSFCM and FKPFCM are faster than RMTPA, ILKFCM, and ALFCM because nonlocal mean filtered image in NSFCM is computed in advance without iteration, and nonkey pixels in FKPFCM are segmented according to the segmentation results of key pixels rather than iterative FCM. FWCRF and THFCM are faster than other algorithms except for CSHLC. In essence, they all deal with pixel blocks in image segmentation. The proposed algorithm is much faster than the other seven comparison algorithms because the K-means used is a fast-clustering algorithm, and MRF has fewer iterations due to the efficient ECS.

IV. CONCLUSION

This article proposes an SAR image segmentation algorithm based on CSHLC. Image edges are used as constraints to conduct Gaussian smoothing on the image. This approach enables the noise in SAR images to be well suppressed while also ensuring that the information of small targets is not destroyed. K-means clustering of the smoothed image is used to initialize an iterative optimization of pixel labels, posed as an MRF. This process corrects erroneous corner labels. Two label correction methods, PGCC and GSC, are used to correct any remaining erroneous labels to further improve accuracy.

CSHLC has been tested in empirical experiments, using both simulated and real SAR images, featuring different sizes and noise levels. The performance of CSHLC has been compared against seven other state-of-the-art SAR image segmentation algorithms from the literature. The experimental results suggest that the CSHLC yields significant advantages in segmentation accuracy, with a particularly strong performance in preserving the corners and small targets. In the simulated SAR image experiments, the CSHLC achieves an overall segmentation accuracy, over all images, above 99%. For high-look images, the accuracy is very close to 100%. This is significantly better than any of the seven comparison algorithms. CSHLC demonstrates state-of-the-art performance on both complex and simple images and, at the same time, has good robustness to noise. In experiments with real SAR images, the segmentation accuracy and kappa coefficient of the CSHLC are significantly higher than the comparison algorithms on the Noerdlinger and Maricopa images. On the Traunstein image, CSHLC also achieves a good segmentation result. These results suggest that CSHLC can effectively segment real SAR images. In addition, the running time comparison experiment, on real images, suggests that CSHLC is significantly faster than the comparison algorithms, indicating high computational efficiency.

CSHLC demonstrates effective segmentation of most small targets and corners in SAR images. However, there are some small targets for which the width is significantly smaller than the minimum smoothing template width. A good example of this is the green segmentation line of the yellow rectangular region in the middle of the Noerdlinger image, which has a width of only one pixel. For these extremely small targets, CSHLC needs further improvement. In future work, we will focus on retaining and correcting such extremely small targets to further improve the accuracy of the algorithm. In addition, we will consider using adaptive K-means instead of fixed K-means to study their effect on segmentation accuracy and efficiency.

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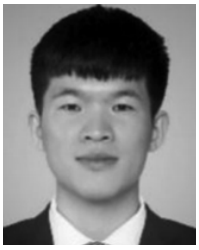
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